Try these

1. Observe the Fig. 2.14 and answer the following questions.

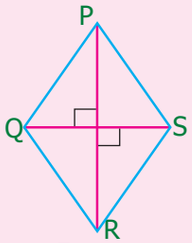


Fig.2.14

(i) Name two pairs of opposite sides.
(ii) Name two pairs of adjacent sides.
(iii) Name the two diagonals

2. Find the area of the rhombus given in Fig. 2.15 and Fig. 2.16

(i)

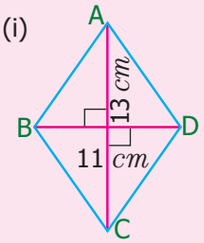


Fig.2.15

(ii)

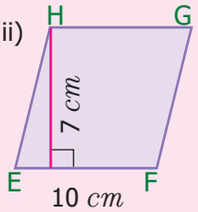


Fig.2.16

Example 2.6

Find the area of the rhombus whose side is 17 cm and the height is 8 cm.

Solution

Given:

Base = 17 cm, height = 8 cm

Area of the rhombus = b x h $sq.\ units$

= 17 x 8 = 136

Therefore, area of the rhombus = 136 $sq.\ cm.$

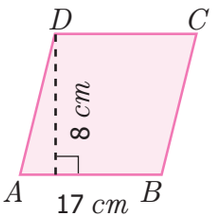


Fig.2.17

Think

1. Can you find the perimeter of the rhombus?
2. Can diagonals of a rhombus be of the same length?
3. A square is a rhombus but a rhombus is not a square. Why?
4. Can you draw a rhombus in such a way that the side is equal to the diagonal.

Example 2.7

Calculate the area of the rhombus having diagonals equal to 6 m and 8 m.

Solution

Given: d_1 = 6 m, d_2 = 8 m

Area of the rhombus = $\frac{1}{2}$ x (d_1 x d_2) $sq.\ units$

= $\frac{1}{2}$ x (6 x 8)

= $\frac{48}{2}$

= 24 $sq.m$

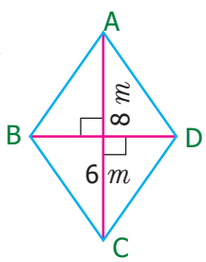


Fig.2.18

Hence, area of the rhombus is 24 $sq.m.$

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